

TAM THUC STRATEGEM

Deployment, CI/CD and Infrastructure Detailed v1.0

1. CI/CD PIPELINE

Sử dụng **GitHub Actions**:

- Lint + Type check + Unit tests
- Build Docker images
- Security scanning (Trivy/Snyk)
- Deploy to Staging (auto)
- Deploy to Production (manual approval gate)

Công cụ: GitHub Actions + Docker + Kubernetes (hoặc Render/Fly.io cho MVP đơn giản hơn)

2. INFRASTRUCTURE

- **Backend**: FastAPI trên Kubernetes hoặc Platform as a Service (Render, Fly.io, Railway)
- **Database**: Managed PostgreSQL (Supabase, Neon, hoặc AWS RDS)
- **Vector DB**: Pinecone hoặc Chroma (self-hosted)
- **Cache**: Redis
- **Storage**: S3 hoặc equivalent cho reports
- **Monitoring**: Prometheus + Grafana + Sentry

3. DISASTER RECOVERY

- Daily automated backups cho database
- Point-in-time recovery
- Multi-AZ deployment
- Documented runbook cho failover